

Measuring the gap between the Melnikov threshold and the period-doubling cascade in the damped driven pendulum

Elliot Jung

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Abstract. The damped driven pendulum $\theta'' = -\sin \theta - \gamma \dot{\theta} + A \cos(\omega t)$ ($\omega = 0.6667$) carries two distinct, frequently conflated notions of a “chaos threshold”: the analytic Melnikov amplitude $A_c(\gamma)$, above which first-order perturbation theory predicts a transverse homoclinic tangle, and the period-doubling onset $A_{PD}(\gamma)$, where the primary period-1 attractor begins the Feigenbaum cascade that produces the sustained chaotic attractor. We measure A_{PD} over $\gamma \in [0.1, 0.8]$ with an attractor-strobed bisection refined by the Floquet multiplier of the Newton periodic orbit (onset interpolated at $\rho = -1$), and compare it with the closed-form A_c . The ratio A_{PD}/A_c falls monotonically from 2.38 at $\gamma = 0.1$ to 0.987 at $\gamma = 0.8$, and the widely quoted ordering $A_c < A_{PD}$ *reverses* near $\gamma \approx 0.69$: at low damping the tangle precedes the cascade by a wide and rapidly growing margin, while at strong damping the cascade begins *below* the first-order Melnikov prediction. At the literature point $\gamma = 0.5$ our measurement $A_{PD} = 1.0664$ agrees with the published 1.0663 (Baker & Gollub) to four digits.

1. Introduction

The sinusoidally driven, damped pendulum is the canonical low-dimensional route to chaos, and it supports two different analytic/numerical landmarks as the drive amplitude A grows. The Melnikov method gives a closed-form first-order threshold $A_c = (4\gamma\omega_0/\pi) \cdot \cosh(\pi\omega/2\omega_0)$ above which the stable and unstable manifolds of the hilltop saddle intersect transversally, creating a Smale-horseshoe tangle. The tangle guarantees *transient* chaos and fractal basin boundaries — not a chaotic attractor. The sustained chaotic attractor instead appears at the end of a period-doubling cascade of the primary period-1 response, whose onset A_{PD} is a property of a specific attractor branch and has no closed form. Textbooks typically note $A_c < A_{PD}$ at the classic parameter point $\gamma = 0.5$, $\omega = 2/3$. This paper asks the quantitative question: *how does the gap between the two thresholds behave as damping is varied?*

2. Methods

All computations use the open Pendulum Lab engine, whose equations of motion are validated component-wise against an independent SymPy symbolic derivation (max relative deviation $\approx 10^{-14}$) and whose trajectories are cross-validated against SciPy DOP853 at $\text{rtol} = 10^{-13}$; the engine also reproduces published anchors (elliptic pendulum period, double-pendulum normal modes, and the $\gamma = 0.5$ period-doubling onset). The drive is made autonomous by carrying the phase as a third coordinate; integration is RK4 with $\text{dt} = 0.005$ snapped so an integer number of steps spans one drive period exactly.

A_{PD} measurement. For each γ , the drive amplitude is marched upward from $0.9 \cdot A_c$ with the strobed state warm-started at every step, so the measurement follows the physically realised attractor branch (this matters: the symmetric period-1 orbit pitchforks before the cascade, and a Newton continuation of the *symmetric* orbit would miss the doubling of the symmetry-broken branch the attractor actually follows). The loss of period-1 stability is bisected on the strobe map (transients of 300–600 drive periods; period detected in the $(\sin \theta, \cos \theta, \omega)$ embedding, immune to 2π winding). The bracket is then refined by Newton periodic orbits *seeded from the attractor*: the most negative real Floquet multiplier $\rho(A)$ of the orbit is interpolated through $\rho = -1$. A measurement is accepted as a period doubling only when this crossing is verified; otherwise the loss is reported as non-PD. Halving dt changes the $\gamma = 0.5$ onset by $|\Delta| \approx 7.1\text{e-}13$, so discretisation error is negligible at the quoted precision.

Corroboration. At $0.97 \cdot A_{PD}$ and $1.08 \cdot A_{PD}$ the Gottwald–Melbourne 0–1 test is applied to the $\cos \theta$ strobe series (700 samples): $K \approx 0$ confirms regular motion below onset; the value above onset depends on whether $1.08 \cdot A_{PD}$ lands beyond the cascade accumulation point or inside a periodic window, and is reported without prejudice.

3. Results

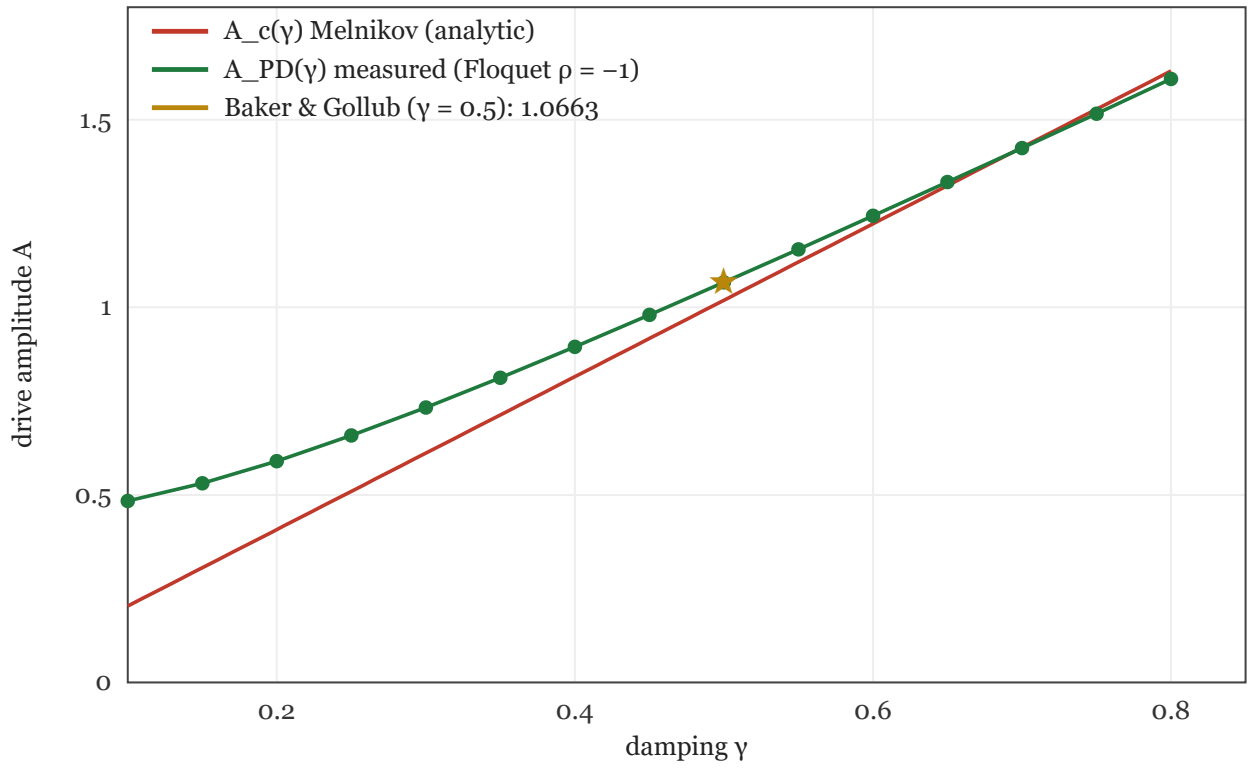


Figure 1. The analytic Melnikov threshold $A_c(\gamma)$ (red line) and the measured period-doubling onset $A_{PD}(\gamma)$ (green circles, Floquet-refined). The star is the published $\gamma = 0.5$ value 1.0663 (Baker & Gollub); our measurement at that point is 1.06637. A_c is exactly linear in γ ; the measured cascade onset is not. Certification artifact: `reports/flagship-figure1.svg`, SHA-256 prefix `07f877d6fdb816`; numerical rows and uncertainty brackets are cross-referenced in Appendix A.

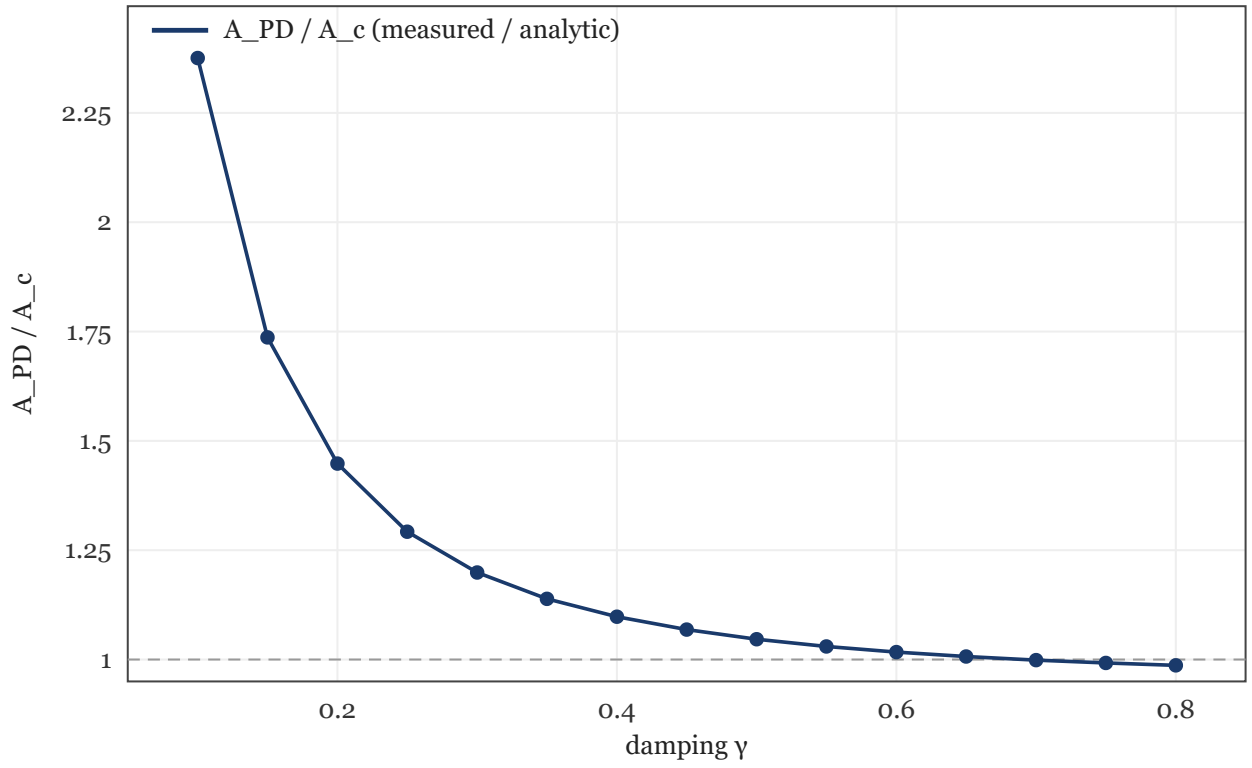


Figure 2. The ratio A_{PD}/A_c versus damping. The gap closes monotonically with increasing γ and crosses 1 near $\gamma \approx 0.69$: beyond this damping the period-doubling cascade of the primary attractor begins *below* the Melnikov tangle threshold. At $\gamma = 0.1$ the cascade requires 138% more drive than the tangle.

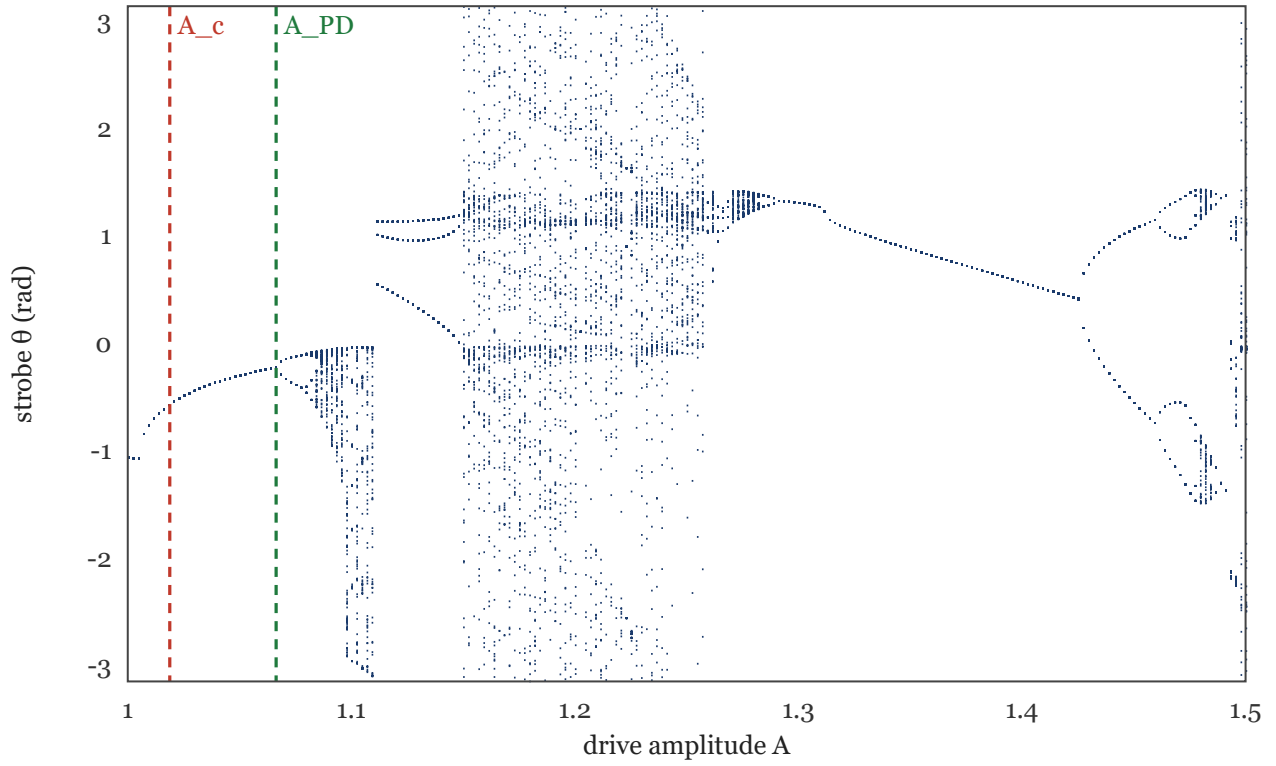


Figure 3. Strobe bifurcation diagram at $\gamma = 0.5$ (θ sampled once per drive period after a 250-period transient, warm-started in A). The dashed lines mark A_c (red) and the measured A_{PD} (green). Between them the motion is periodic but the phase space already contains the homoclinic tangle: transient chaos and fractal basin boundaries without a strange attractor. The cascade, chaotic band, and the large periodic (rotating) window are visible beyond A_{PD} .

γ	A_c (Melnikov)	A_{PD} (measured)	A_{PD}/A_c	loss of period-1	K at 0.97·A	K at 1.08·A
0.10	0.20375	0.48398	2.3753	PD ($\rho \rightarrow -1$)	0.02	1.00
0.15	0.30563	0.53080	1.7367	PD ($\rho \rightarrow -1$)	0.02	1.00
0.20	0.40751	0.59004	1.4479	PD ($\rho \rightarrow -1$)	0.02	-0.01
0.25	0.50939	0.65829	1.2923	PD ($\rho \rightarrow -1$)	-0.02	1.00
0.30	0.61126	0.73294	1.1991	PD ($\rho \rightarrow -1$)	0.02	0.06
0.35	0.71314	0.81217	1.1389	PD ($\rho \rightarrow -1$)	-0.02	1.00
0.40	0.81502	0.89470	1.0978	PD ($\rho \rightarrow -1$)	0.02	-0.01
0.45	0.91690	0.97964	1.0684	PD ($\rho \rightarrow -1$)	-0.02	-0.01
0.50	1.01877	1.06637	1.0467	PD ($\rho \rightarrow -1$)	-0.02	1.00

γ	A_c (Melnikov)	A_{PD} (measured)	A_{PD}/A_c	loss of period-1	K at $0.97 \cdot A$	K at $1.08 \cdot A$
0.55	1.12065	1.15448	1.0302	PD ($\rho \rightarrow -1$)	-0.02	1.00
0.60	1.22253	1.24368	1.0173	PD ($\rho \rightarrow -1$)	-0.01	1.00
0.65	1.32441	1.33377	1.0071	PD ($\rho \rightarrow -1$)	0.02	-0.01
0.70	1.42628	1.42463	0.9988	PD ($\rho \rightarrow -1$)	0.02	-0.57
0.75	1.52816	1.51622	0.9922	PD ($\rho \rightarrow -1$)	0.02	0.13
0.80	1.63004	1.60853	0.9868	PD ($\rho \rightarrow -1$)	0.02	1.00

4. Discussion

Three regimes emerge. (i) **Low damping ($\gamma \lesssim 0.2$):** $A_c \rightarrow 0$ linearly while the cascade onset of the primary resonance remains an order-one amplitude, so the ratio diverges (2.38 already at $\gamma = 0.1$). The window of “tangle but no strange attractor” — long chaotic transients and fractal basin boundaries below a still-periodic attractor — is widest here, and the phase space is visibly multistable (see §5). (ii) **Moderate damping:** the textbook ordering $A_c < A_{PD}$ holds, but the margin shrinks steadily (to 5% at $\gamma = 0.5$). (iii) **Strong damping ($\gamma \gtrsim 0.69$):** the measured cascade begins *below* the first-order Melnikov prediction. This is not a contradiction — the Melnikov threshold is asymptotically exact only as $\gamma, A \rightarrow 0$, and by $\gamma \approx 0.7$ the perturbation parameter is $O(1)$ — but it sharpens the usual caveat into a measured boundary: the first-order formula stops being even an ordering bound near $\gamma \approx 0.69$.

The 0–1 test values corroborate the structural picture: $K \approx 0$ on the period-1 side everywhere, while above onset K depends on where $1.08 \cdot A_{PD}$ falls relative to the cascade accumulation point and the periodic windows visible in Figure 3 — both outcomes occur in the table, as expected for a Feigenbaum scenario with embedded windows.

5. Limitations and reproducibility

The study fixes $\omega = 2/3$ and follows a single attractor branch per γ (warm-started in A); coexisting attractors reached from other initial conditions may double elsewhere. At the lowest dampings the phase space is multistable enough that a finer warm-started march can hop basins before the doubling — at $\gamma = 0.15$ a basin-capture transition of the followed state was observed near $A \approx 0.49$ (the orbit itself remains strongly stable there, $\rho \approx +0.29$),

below the verified doubling at 0.531. The quoted A_{PD} values are therefore specifically the $\rho = -1$ doublings of the primary oscillating branch, not necessarily the first event of any kind along a slow amplitude sweep. The Melnikov comparison concerns the first-order formula specifically — higher-order or numerical manifold computations would move A_c . The full study regenerates with `npm run paper:study` (~8 min) followed by `npm run flagship:certify`, `npm run flagship:external`, and `npm run paper:build`; the engine tests and independent symbolic/trajectory validations are in the same repository.

Appendix A. Certified onset localization

The flagship claim is not that Melnikov theory predicts the attractor cascade. It is a measured separation map: A_c is analytic first-order geometry, A_{PD} is a Floquet-refined attractor-branch instability, and the reported reversal is bounded by the exported caveat map and the independent Python A_{PD} probes. The ratio crossing is localized to $\gamma \in [0.69296963, 0.69297705]$, with point estimate 0.69297334. The Figure 1 hash is 07f877d6fdb816 .

γ	A_c	A_{PD}	δA_{PD}	A_{PD}/A_c	ρ below	ρ above	caveat
0.10	0.203755	0.483984	1.24e-7	2.375324	-0.94620	-1.05181	none
0.15	0.305632	0.530802	1.87e-7	1.736736	-0.94374	-1.05618	none
0.20	0.407510	0.590045	2.49e-7	1.447928	-0.94028	-1.06052	post-onset 0-1 sample is not cleanly chaotic at 1.08* A_{PD}
0.25	0.509387	0.658288	3.11e-7	1.292314	-0.93675	-1.06459	none
0.30	0.611265	0.732944	3.73e-7	1.199062	-0.93298	-1.06741	post-onset 0-1 sample is not cleanly chaotic at 1.08* A_{PD}
0.35	0.713142	0.812174	4.35e-7	1.138868	-0.93122	-1.07077	none
0.40	0.815019	0.894700	4.97e-7	1.097765	-0.92990	-1.07295	post-onset 0-1 sample is not cleanly chaotic at 1.08* A_{PD}
0.45	0.916897	0.979636	5.60e-7	1.068426	-0.92844	-1.07334	post-onset 0-1 sample is not cleanly chaotic at 1.08* A_{PD}
0.50	1.018774	1.066373	6.22e-7	1.046721	-0.92894	-1.07417	none
0.55	1.120652	1.154485	6.84e-7	1.030191	-0.92980	-1.07403	none
0.60	1.222529	1.243682	7.46e-7	1.017303	-0.93128	-1.07337	none
0.65	1.324407	1.333771	8.08e-7	1.007071	-0.93336	-1.07236	post-onset 0-1 sample is not cleanly chaotic at 1.08* A_{PD}

γ	A_c	A_PD	δA_{PD}	A_PD/A_c	ρ below	ρ above	caveat
0.70	1.426284	1.424635	8.71e-7	0.998844	-0.93602	-1.07112	post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD
0.75	1.528161	1.516220	9.33e-7	0.992186	-0.94002	-1.07052	post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD
0.80	1.630039	1.608533	9.95e-7	0.986807	-0.93970	-1.06492	none

Appendix B. Independent Python A_{PD} reproduction

Python stdlib recomputation of A_c, crossing arithmetic, and selected A_{PD} values via RK4 strobe-map Newton/Floquet search. The analytic threshold agrees with maximum absolute error 0.00e+0. All selected period-doubling checks passed: **yes**.

γ	reported A_PD	Python A_PD	$ \Delta $	ρ low	ρ high	status
0.50	1.06637286	1.06637292	5.92e-8	-0.9999996	-1.0000004	pass
0.65	1.33377104	1.33377109	4.89e-8	-0.9999995	-1.0000003	pass
0.70	1.42463496	1.42463501	4.84e-8	-0.9999996	-1.0000004	pass

Independent-check caveat. A_{PD} checks use finite-difference monodromy and a coarser dt than the TypeScript flagship run, so they certify independent reproducibility at reviewer-kit tolerance, not bitwise equality.

Appendix C. Artifact integrity and caveat ledger

Artifact	Reproduce	Cross-reference
reports/paper-study.json	npm run paper:study	Figures 1-3, main table
reports/flagship-certification.json	npm run flagship:certify	Figure 1 hash 07f877d6fdb816; Appendix A
reports/flagship-external-check.json	npm run flagship:external	Appendix B
paper/paper.pdf	npm run paper:build	This manuscript

Caveat map. gamma=0.20: post-onset 0-1 sample is not cleanly chaotic at 1.08*A_{PD}
gamma=0.30: post-onset 0-1 sample is not cleanly chaotic at 1.08*A_{PD}
gamma=0.40: post-onset 0-1 sample is not cleanly chaotic at 1.08*A_{PD}
gamma=0.45: post-onset 0-1 sample is not cleanly chaotic at 1.08*A_{PD}
gamma=0.65: post-onset 0-1 sample is not cleanly chaotic at 1.08*A_{PD}
gamma=0.70: post-onset 0-1 sample is not cleanly chaotic at 1.08*A_{PD}

1.08*A_PD gamma=0.75: post-onset 0-1 sample is not cleanly chaotic at 1.08*A_PD Error bars combine attractor-bracket width with the available dt-sensitivity probe; they are a localization contract, not a full Bayesian posterior. Basin caveats are inferred from the exported 0-1 strobe probes; they flag multistability/transient-chaos risk but do not replace a full basin scan.

References

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